

E_Lecture5_W2

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Q1 [20 pts, 10 pts for each]

亮亮在釣魚台島上釣魚，在使用一套漁具之下，其釣魚時數(L)與漁獲量

(STP) 之間關係如下表所示:

Alex went fishing on the island. He used a fishing tackle. The time he took is L and the corresponding yield of catch is TP. The relation between L and STP is as follows.

L(hr)	1	2	3	4	5	6
STP	5	15	22	28	32	33

(1)

試由表中的數據，推導亮亮勞動力投入的平均產量、邊際產量

Calculate the SAP_L and SMP_L according to the table.

$$SAP_L = \{STP_L\}/L$$

$$SMP_L = \{STP_L - STP_{\{L-1\}}\} / \text{change of } L$$

L	SAP	SMP
1	5	5
2	7.5	10
3	7.33	7
4	7	6
5	6.4	4
6	5.5	1

(2)

假設每套漁具每 6 個小時的租金為 100 元，而亮亮每小時可賺得的最高

薪資為 300 元。試由上表所給的數據推導總成本、總固定成本、總變動

成本

If the rent of the fishing tackle is \$100 per 6hrs, and the highest salary Alex can obtain per hours is \$300. Calculate the TC, TFC, and TVC according to the table. $TFC_0 = TFC_1 = \dots = TFC_6 = 100$ $TVC = wL = 300L$ $TC = TVC + TFC = 100 + 300L$				
	L	TC	TFC	TVC
	1	400	100	300
	2	700	100	600
	3	1000	100	900
	4	1300	100	1200
	5	1600	100	1500
	6	1900	100	1800

Q2

[20 pts, 10 pts for each]

健荃在街上賣枝仔冰，其固定成本為 20 元，冰棒的數量(STP)與總成本

(STC) 的關係如下表

Wendy is the seller of ice cream. In the short run, her fixed cost of selling ice cream is \$20. The relation between her quantity supplied (STP) and the corresponding total cost (STC) is as follows.

STP (Units)	1	2	3	4	5	6	7
STC (\$)	20	30	38	45	50	56	65

(1)

試由表中的數據推估短期之內各產量對應的總固定成本(STFC)、總變動成本

(STVC)、平均固定成本(SAFC)、平均變動成本(SAVC)、短期平均成本(SAC)、

邊際成本(SMC)

With the information above, calculate the STFC, STVC, SAFC, SAVC, SAC and SMC

$STFC_n - STFC_{n-1} = 0, STFC_0 = STFC = \dots = STFC_7 = 20$

$$STVC_n = STC - STFC = STC - 20$$

$$SAFC_n = STFC / n$$

$$SAVC_n = STVC / n$$

$$SAC_n = STP / n = SAFC + SAVC$$

$$SMC_n = SMC_n - SMC_{\{n-1\}}$$

STP	STFC	STVC	SAFC	SAVC	SAC	SMC
1	20	0	20	0	20	0
2	20	10	10	5	15	10
3	20	18	6.67	6	12.67	8
4	20	25	5	6.25	11.25	7
5	20	30	4	6	10	5
6	20	36	3.33	6	9.33	6
7	20	45	2.86	6.43	9.29	9

(2)

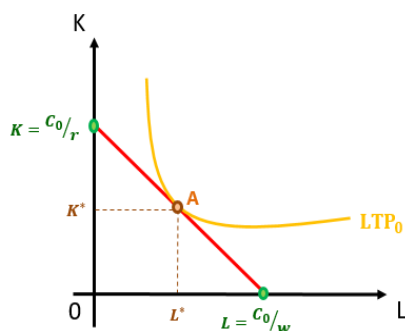
如果每一枝枝仔冰賣 10 元, 在健荃賣出 3 枝冰棒的情況下, 利潤為多少?

Suppose that the price of a ice cream is \$10. What is her profit if she sells three ice cream?

$$\text{Profit} = 30 - 38 = -8 \text{ (Loss)}$$

Q3

[30 pts]



- 左圖中，如果每單位的資本租賃費用 r 不變，但工資率 w 上升至 w' ，如果租賃與勞工的總使用單位不變，則總成本水準將從 C_0 提升至 C_1

假定 $\frac{C_0}{w} = \frac{C_1}{w'}$ ，則如果要維持同樣的總產量 LTP_0 ，廠商的最適要素組合(給定總產量，總成本最低的要素組合)將如何變動?

- 長期總成本(LTC)又將如何變動?

- In the above figure, we suppose that the cost of using one unit of fixed factor is r and the cost of using one unit of variable cost is w . Once

	the w rises to w' and the r is unchanged so that the total cost increases from C_0 to C_1. If $\frac{C_0}{w} = \frac{C_1}{w'}$, what is the producer's input bundle having the lowest Long-Term Cost (LTC) given that the its total product LTP_0 is unchanged. - What's the corresponding LTC if the producer chooses the optimal input bundle? a. According to the substitute effect, when the labor fees(variable cost v) rise will increase the input of capital(K) and decrease the input of labor(L). The w/r will become steeper b. The range of LTC is between C_0 and C_1. LTC must bigger than C_0 as w rises (w') will affect the cost higher than before. A rational firm will change the optimal point to product same number of product while using minimum cost, it will let LTC lower than C_1.
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Q4	[30 pts, 10 pts for each] (1) A firm is producing 20 units with an average cost (SAC or LAC) of \$25 and a marginal cost (SMC or LMC) of \$15. If it were to increase production to 21 units, which of the following must occur? (a) Marginal cost would decrease. (b) Marginal cost would increase. (c) Average cost would be decrease. (d) Average cost would be increase. Answer: ___C___ (2) The government imposes a \$1000 per year license fee on all pizza restaurants. As a result, which cost curves shift? (a) average cost (SAC or LAC) and marginal cost (SMC or LMC) (b) average cost and average fixed cost (SAFC or LAFC).
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(c) average variable cost (SVAC or LAVC) and marginal cost

(d) average variable cost and average fixed cost.

Answer: ____B____

(3)

If a higher level of production allows workers to specialize in particular tasks, a firm will likely exhibit _____ of scale and _____ average cost (LAC).

(a) economies, falling

(b) economies, rising.

(c) diseconomies, falling.

(d) diseconomies, rising.

Answer: ____a____